

Ch8_Data ethics and legal perspectives

一、Introduction to Data Ethics

1. Definition and importance

Data ethics encompasses 包含 the moral obligations of gathering, protecting, and using personally identifiable information and how it affects individuals.

- Protect information: 采取技术和制度, 确保个人数据在收集、存储、处理和共享过程中不被泄露 leakage、滥用 misuse 或非法访问 Unauthorized access

Data ethics refers to the **moral considerations** surrounding data collection, analysis and use.

They emphasise the importance of upholding ethical standards to promote trust, fairness and integrity in data-driven decision-making processes.

1.1. Importance

1. Protection of individual rights

Without robust ethical guidelines, there's a risk of [infringing upon individual rights to privacy](#), autonomy and dignity.

2. Mitigation of bias and discrimination

Unethical data practices, such as biased algorithms or discriminatory profiling, may result in unfair treatment or discrimination against certain groups.

3. Enhancement of trust and transparency

Transparency builds confidence in data processes and ensures accountability for the consequences of data-driven decisions.

4. Safeguarding against harm

5. Promoting social responsibility

在当今高度互联的世界 increasingly interconnected world 里, 数据伦理不仅仅是 individual organisations 企业内部的合规问题, 它还涉及到 整个社会的影响 broader societal impacts。

- 企业在收集、使用和保护数据时, 不仅要考虑自身利益, 还要考虑对 个人、群体、社区和社会整体的影响。
- 当数据实践遵循伦理原则时, 它能促进社会责任, 推动 positive social outcomes 和 sustainable development

2. Key ethical principles

- **Fairness** – 不主观臆断；数据使用和算法决策不能带有偏见或歧视
- **Accountability** – 权责分明：数据的收集、处理和使用必须有明确的责任归属，出问题时能追责
- **Transparency** – How to train the data? 、算法逻辑、模型训练方式应对用户或监管机构可解释； 减少“黑箱操作”；
- **Privacy**
- **Consent** – Seek explicit permission from individuals before collecting their personal data – 尊重知情权

Key aspects of data ethics include:

- Respecting individual rights to **privacy** and autonomy
- Obtaining informed **consent** 同意 for data collection and use
- Being **transparent** about data practices
- Mitigating the risk of harm or **discrimination** 歧视

3. **Ethical Risk** Categories

Table 9.1: Ethical risks in data-driven decision making				
Bias	Discrimination	Manipulation	Exploitation	Harm

3.1. Bias -> Discrimination

- Data, algorithms -> 系统性偏差 -> Unfair
- 指在数据收集、处理或算法设计过程中产生的系统性偏差，导致结果不准确或不公平。
 - 如果训练数据主要来自城市用户，模型可能对农村用户预测不准确。

【对比 偏向 (Skewed Outcomes / Favoritism) 】

分析结果或决策逻辑在利益分配上更有利于某些群体，即使过程本身没有明显技术偏差。

3.2. Discrimination

- 对某些群体造成不公正待遇。

概念	Bias	Discrimination
层次	数据或模型中的偏差	决策或结果中的不公平
性质	多为无意、技术性	多为结果性，有时带制度性
表现	算法预测不准确	某群体被不公平对待

3.3. Manipulation (操控)

- 利用数据影响用户行为，使其做出**非自愿或不知情的**决策。
 - 社交平台通过算法推送特定内容，引导用户情绪或消费。

3.4. Exploitation (剥削)

- **利用用户数据牟利**，而不考虑用户权益或知情权。
- 指组织或个人在使用数据时，**过度或不公平地**利用用户信息来牟利，而忽视 Ignoring users' privacy, or right to be informed.
 - 企业收集用户数据后，未经用户同意就出售给第三方

3.5. Harm (伤害)

- 数据使用直接或间接导致用户受到心理、经济或社会伤害。
 - 错误的健康推荐导致用户延误治疗；
 - 算法误判导致失业。

二、 Legal and Regulatory Landscape

The "legal and regulatory landscape" refers to the **complete set of laws, regulations, and policies** that **govern the operations** of individuals and businesses within a specific industry or region.

landscape:

Legal Risks in Data Usage

Emerging Global Trends in AI Regulation

三、 **Social Risks** and implications

1. Overview

影响:

- 人类尊严 (Human dignity)
- 自主权 (Autonomy)
- 公平正义 (Justice)
- 民主机制 (Democracy)

目标: Balancing **benefits** and **risks** for(为了) the common good 公共利益

2. Category

- **Privacy**

- Misuse or unauthorized access to data
- Who can use? How to control?
- **Protective measures** 保护性措施 : Data minimization 数据最小化, Encryption, Access control, Informed consent
 - Data Minimization (数据最小化) => Collecting only the data necessary for a specific purpose.
 - Reduces risks of misuse, breaches, and privacy violations by limiting unnecessary data storage.

- **Discrimination**

- 不公平对待
- **Mitigation strategies**: 公平性审查 Fairness Audit、算法去偏 Algorithmic Debiasing 、包容性设计 Inclusive Design
 - Fairness Audit – A systematic evaluation of algorithms or data systems to check whether they treat different groups fairly.
 - Algorithmic debiasing – Techniques used to reduce or eliminate bias in algorithms and datasets.
 - Inclusive design – Ensures products, systems, and services are accessible and usable by diverse populations.
 - It emphasizes considering marginalized or **vulnerable** groups during design, not just the majority.

- **Manipulation**

- 风险类别, 指算法或数据在**不透明**的情况下影响用户行为、观点或选择。
- Personalized recommendations, filter bubbles
- **Countermeasures** 对策: Algorithmic transparency 算法透明; User control 用户控制权; Content diversity 信息多样性保障

- **Accountability**

- 核心问题是: 当算法出错或造成伤害时, **谁负责**?
- **Solutions**: Explainable design 可解释性设计, **responsibility allocation** 责任分配机制, legal frameworks 法律框架

- Explainable design 解决方案，指让算法的决策过程对人类可理解

Algorithmic Transparency	公开算法逻辑和推荐原因	Manipulation（操控）	推荐系统、广告投放
Explainable Design	让 <u>决策过程可理解</u> 、 <u>责任可追溯</u>	Accountability（问责）	信贷审批、医疗诊断、司法判决
